

Evaluation of BTA-TKPP Corrosion Inhibition for Mild Steel in Desalination Pipelines

Isaac Summers
Student Number: 492991

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Tutor: Dr Stefano Mori

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1. Aim, real-world context and reasoning

The global desalination industry relies on carbon steel pipelines to transport highly saline seawater and brine, where chloride concentrations accelerate corrosion by disrupting the protective oxide layer on the metal surface. This leads to wall thinning, pitting, leaks, unplanned maintenance, safety hazards and substantial economic losses (Shokri & Sanavi Fard, 2022).

The aim of this work is to evaluate the effectiveness of a corrosion inhibitor for mild steel in high salinity environments. Inhibitor performance is quantified as a function of concentration using daily electrochemical (EC) measurements from open circuit potential (OCP), linear polarisation resistance (LPR) and electrochemical impedance spectroscopy (EIS) methods. Film formation, stability and corrosion impact will also be analysed through EC trends and through post exposure tensile testing and SEM/EDX microscopy. The findings will help inform corrosion control strategies for desalination pipelines exposed to saline environments.

It is hypothesised that the inhibitor mitigates corrosion by adsorbing onto the steel surface to form a protective film, with effectiveness increasing with inhibitor dosage.

2. Experimental Methodology

2.1. Materials and Test Conditions

Mild steel dog-bone specimens were tested in a 3.5 wt.% NaCl solution at 15°C, simulating seawater. The inhibitor used was benzotriazole-tetrapotassium pyrophosphate (BTA-TKPP), dosed at 0 (uninhibited), 50 (low dose) and 500 ppm (high dose) to reflect typical literature ranges showing dose-response behaviour (Ali et al., 2024; Mohammed et al., 2022).

Prior to immersion, the gauge section was ground for reproduceable surface finish, rinsed with deionised water, degreased with acetone, rinsed again and dried in air. Tests used a 250ml beaker with 200 ml solution.

2.2. Electrochemical Set-Up

A three electrode cell was employed (depicted in Appendix A, Figure A1(a)): working electrode (dogbone gauge section), Ag/AgCl reference, platinum counter. Measurements (OCP, LPR, EIS) were performed daily (days 1-4, 7) using PalmSens4 and PSTrace 5.9 for data visualisation. While the setup doen't strictly comply with a formal standard, methods such as NACE TM0284 or ASTM G106/G59 could have been followed to ensure controlled temperature and standardised electrode arrangement.

Technique	Parameters	Standards Used
OCP	120 seconds total, 0.1 s interval	-
LPR	±10 mV, cyclic polarisation	ASTM G102 (corrosion rate calculation)
EIS	10 mHz – 100 kHz	ISO 16773-2 (frequency range)

***Table 1**: Electrochemical test parameters*

2.3. Post Exposure Testing

Due to time constraints, tensile tests conducted on day 7 were done at 5 mm/min rather than as per ASTM E8M standards, with setup show in Appendix A, Figure A1 (b).

Metallography (cross section to fracture): hot mounted in a 30mm carbon filler resin cylinder at 195°C, ground using 240-2500 SiC, followed by polishing with 9-3 µm diamond suspensions, final polish with 1 µm silica and etched in 2% nital for 10s to reveal the microstructure.
Imaging: Macroscopy using Zeiss Axio Vert.A1 to asses pitting/film uniformity, microscopy using Zeiss

Axio Observer to evaluate grain attack. SEM-EDX with TESCAN VEGA3 (BSE mode) for 1-100 µm morphology and Fe/O/C mapping.

3. Results

3.1. Electrochemical Response

OCP measurements revealed distinct trends across samples (Appendix B, Figure B1 (a)). For the uninhibited sample (0 ppm), potential dropped on days 1-2 due to natural oxide dissolution, followed by a noisy rise (Appendix B, Figure B1 (b)) indicating porous corrosion product accumulation and active corrosion (Jakeria et al., 2022). At 50 ppm, potential increased in days 1-4 (film formation) with low noise, then declined, suggesting insufficient long term protection. At 500 ppm, potential stabilised at higher (more noble) values by day 7, indicating formation of a more stable protective film compared with the lower inhibitor dose (Jakeria et al., 2022).

LPR-derived corrosion rates (Appendix C, Figure C1 (a); shows methodology as per ASTM G102, with constants (density, equivalent weight, Stern Geary) from (Baboian, 2002)) averaged 0.109 mm/y for the uninhibited sample by day 7, with 61% inhibition at 50 ppm and 39% at 500 ppm. These corrosion rates are within typical magnitudes for mild steel in seawater (Baboian, 1995). Despite being a higher dose, 500 ppm is less effective overall, consistent with literature in which excess inhibitor causes molecular clustering forming imperfect films vulnerable to Cl^- attack (Teymouri et al., 2021).

Dosage (ppm)	Corrosion Rate (mm/yr)					Avg Inhibitor Efficiency	Day 7 Inhibitor Efficiency
	Day 1	Day 2	Day 3	Day 4	Day 7		
0	0.094	0.118	0.121	0.095	0.117	N/A	N/A
50	0.016	0.043	0.044	0.049	0.061	~ 61%	~ 47%
500	0.105	0.057	0.049	0.054	0.070	~ 39%	~ 40%

***Table 2**: Corrosion rates for each sample*

EIS data fitted to a Randles circuit, with $\chi 2 \leq 0.13$ for all fits (Appendix D, Figure D1 (a)), showed solution resistance (R_s) ≈ 3-7 Ω consistent across samples and days. R_{ct} increased until day 3 (film compaction at 217 Ω at 50 ppm, 267 Ω at 500 ppm) then declined (95 Ω and 124 Ω by day 6), indicating partial film break down and increased porosity (Jakeria et al., 2022), also aligning with LPR (R_p) confirming both methods capture the same barrier behaviour (Appendix D, Figure D1 (b)).

3.2. Mechanical Response and Microstructure

Tensile testing (method and results table outlined in Appendix E) yielded similar UTS across doses (455-462 MPa, ~1.3% variation attributed to scatter), however 50ppm exhibits a ~13% drop in elongation relative to the 0 ppm, which is unexpected considering its superior inhibition efficiency. This could be due to the inhibitor's protective film constraining near surface plastic deformation and locally promoting more brittle cracking, consistent with observations that brittle surface films on ductile substrates can delay plasticity and induce premature fracture (Guo et al., 2015).

Macroscopy (Appendix F, Figure F1) showed rust coverage decreasing with dose (sparse at 500 ppm) and appears more uniformly distributed at 500 ppm consistent with the more stable OCP behaviour observed, suggesting enhanced inhibitor film stability rather than corrosion suppression alone (Jakeria et al., 2022). SEM-BSE imaging (Appendix F, Figure F1) shows block like features detaching from the steel surface for all conditions. At 0 ppm, these features are smaller and less frequent, consistent with thin corrosion product formation, at 50 ppm and 500 ppm, detached fragments were similar in size but occurred more frequently with increasing inhibitor dose. EDX mapping and point analysis (Appendix F, Figure 1, Table F1) showed the detached features as iron oxide with higher carbon likely due to a thin covering layer of resin, the main steel layer was largely unoxidized (Fe ≈ ~98% wt.%). The increase in fragment density at higher

inhibitor dose suggests thicker/more continuous inhibitor/oxide films form but detach locally rather than forming finely distributed products. This is comparable to film thickening and delamination on Al6061 by (Jakeria et al., 2022), and agrees with earlier OCP observations of a more stable final film at 500 ppm.

Corrosion exposure over 7 days had little effect on bulk tensile strength, indicating that the test duration was too short for significant cross sectional damage or microstructural degradation to happen. However, measurable changes in ductility were observed suggesting corrosion primarily affected the surface rather than the bulk material. This behaviour aligns with the macroscopy observations which visually show dose dependant reductions in rust and surface damage.

4. Data quality, Errors and Interpretation

4.1. Electrochemical Uncertainty

Surface area estimation (Appendix C, Figure C1 (b)) used CAD modelling of the specimen and recreation of the experimental setup to mitigate area overestimation from incorrect mounting angle, showing ~5% discrepancy when calculated manually. This transfers directly to R_p and i_{corr}, overestimating rates by up to 5% however still preserves trends.

Uncertainty in absolute corrosion rates also arises from using a standard NACE/ASTM Stern Geary constant (B) rather than a value determined from the present environment. As $i_{corr} = \frac{B}{R_p}$, any error in B transfers linearly, (Paterlini et al., 2024) reports B values between ~30 and ~60 mV for carbon steel in atmospheric conditions, implying use of a single fixed B value could introduce uncertainty in the order of ~±30% for corrosion rates derived from LPR.

4.2. Mechanical Uncertainty

Modulus measurements are lower than literature values (~80 GPa vs ~200 GPa), likely due to the use of crosshead displacement instead of an extensometer (recommended by ASTM E111), which has been shown to underestimate Young's modulus by ~50%. For comparison, modulus values for other materials under different testing conditions range from 66-230 MPa (Marter et al., 2018).

4.3. Impact on Findings

Absolute rates have uncertainty of ~±35% (area, B), but corrosion rate trends and rankings and film kinetics (peak day 3, then breakdown) hold consistent across OCP/EIS/LPR. Mechanical insensitivity over 7 days is reliable for UTS but uncertain for ductility due to less accurate extension measurements.

5. Risk Assessment Discussion

5.1. Real World Hazards in Desalination Pipelines

A risk assessment for a mild steel desalination pipeline is provided in Appendix G, Figure G1, showing hazards likelihood, consequences and controls. The main degradation mechanisms considered are general corrosion, localised pitting, flow accelerated/erosion-corrosion, under deposit corrosion and loss of inhibitor dosing.

The assessment concludes that flow velocity induced erosion-corrosion is the most hazardous mechanism because high shear stresses and turbulent flow in saline brine can rapidly thin pipe walls, driving leaks and ruptures with significant safety, environmental and economic consequences. This aligns with literature indicating that strong flow accelerates corrosion and erosion-corrosion of carbon steels in sea water (Xu et al., 2021).

5.2. Alignment Between Experiment and Main Hazard

Appendix G, Table G1 summarises how the current experiment maps onto some of the key real world hazards. The static immersion test directly captures chloride-driven electrochemical corrosion which is the fundamental mechanism underlying all desalination corrosion hazards. OCP, LPR and EIS quantify film formation, breakdown kinetics and corrosion rates (Appendices B-D), while spectroscopy validates corrosion morphology (Appendix F). However, it fails to reflect flow accelerated erosion-corrosion, the highest risk operational hazard because no hydrodynamics are imposed. Results thus represent a conservative baseline for static chloride attack but underestimate degradation in practice where flow erodes protective films and accelerates mass loss (Shokri & Sanavi Fard, 2022).

6. Recommendations

6.1. Introduce Hydrodynamic Conditions

To capture the dominant flow induced erosion-corrosion hazard identified in Appendix G, testing should employ a recirculating flow loop in which flow velocity, and therefore wall shear, can be controlled to represent pipeline operation at different velocities. Literature confirms that erosion-corrosion produces steel loss far exceeding the sum of losses from corrosion or erosion alone, the equipment that should be used is a closed loop pump system with a flow meter and pressure sensors, similar in concept to the submerged jet loop used by (Xu et al., 2021).

6.2. Extend Duration with Sacrificial Sampling

Tests should be conducted over 30-90 days with multiple sacrificial specimens removed at intervals (72h, 7d, 30d) for SEM/EDX and mechanical testing, allowing for a time resolved correlation of electrochemistry with microstructure and strength degradation. This addresses current uncertainty in long term film stability and mechanical impact.

6.3. Improve Measurement Precision

Use extensometer for tensile modulus per ASTM E111, which should correct the Young's modulus underestimation and verify elongation results. There should be replicate specimens ($n \geq 3$ per condition) to quantify scatter and statistical significance.

7. Conclusion

The laboratory test successfully quantified chloride driven corrosion and inhibitor performance under static immersion, demonstrating that BTA-TKPP at 50 ppm provides optimal initial protection (83% efficiency day 1) but degrades over time, while 500 ppm despite lower initial efficiency shows higher stability by experiment end. However, the risk assessment (Appendix G, Figure G1) identifies flow field velocity induced erosion-corrosion as the most hazardous threat to desalination pipelines due to rapid, mechanical electrochemical wall loss which partially bypasses inhibitor films under turbulent saline flow.

The current static test does not capture this dominant hazard. Therefore, results underestimate degradation rates and overestimate inhibitor endurance in real world operation. Engineers should not rely on these findings without flow loop validation to specify inhibitor dosing for a desalination plant, or the chosen strategy could fail catastrophically earlier than expected due to film erosion, resulting in unplanned maintenance, shutdowns, brine releases, safety incidents from leaks and economical losses from repairs, lost production and regulatory penalties.

Standards such as ASTM G170 (for flow accelerated corrosion testing) and NACE SP0108 (for corrosion control in seawater systems) require dynamic testing and consideration of multiple corrosion mechanisms which would prevent this gap. Re-evaluating methodology to incorporate suggested changes is needed to translate laboratory tests into reliable pipeline protection.

8. References

Ali, N., Fonna, S., Nurdin, N., Saputra, Y., & Arifin, A. K. (2024). Assessment of Syzygium cumini (Jamblang) fruit extract as an eco-friendly corrosion inhibitor for low-carbon steel in 3.5% NaCl medium. *International Journal of Corrosion and Scale Inhibition*, *13*(1), 223–240. https://doi.org/10.17675/2305-6894-2024-13-1-11

Baboian, Robert. (1995). *Corrosion tests and standards : application and interpretation*. ASTM.

Baboian, Robert. (2002). *Nace corrosion engineer's reference book* (R. S. TRESEDER, Ed.; Third). Natl Assn Of Corrosion. https://studylib.net/doc/27622065/nace-corrosion-engineer-s-reference-book

Guo, T., Qiao, L., Pang, X., & Volinsky, A. A. (2015). Brittle film-induced cracking of ductile substrates. *Acta Materialia*, *99*, 273–280. https://doi.org/10.1016/j.actamat.2015.07.059

Jakeria, M. R., Toh, R. J., Chen, X. B., & Cole, I. S. (2022). Evolution and stability of 2-mercaptobenzimidazole inhibitor film upon Al alloy 6061. *Journal of Applied Electrochemistry*, *52*(6), 1021–1044. https://doi.org/10.1007/s10800-022-01687-w

Marter, A. D., Dickinson, A. S., Pierron, F., & Browne, M. (2018). A Practical Procedure for Measuring the Stiffness of Foam like Materials. *Experimental Techniques*, *42*(4), 439–452. https://doi.org/10.1007/s40799-018-0247-0

Mohammed, M. T., Al-Sieadi, W. N., & Al-jeilawi, O. H. R. (2022). Corrosion inhibitor of carbon steel in 3.5%NaCl solution with Schiff base compounds. *International Journal of Health Sciences*, 57–75. https://doi.org/10.53730/ijhs.v6ns6.9172

Paterlini, L., Brenna, A., Ceriani, F., Gamba, M., Ormellese, M., & Bolzoni, F. (2024). Atmospheric Corrosion of Different Steel Types in Urban and Marine Exposure. *Materials*, *17*(24), 6211. https://doi.org/10.3390/ma17246211

Sabarinathan, C., Muthu, S., & Ali, M. N. (2012). ORIGINAL ARTICLES Experimental Study On Tensile Behavior Of Multi Wall Carbon Nanotube Reinforced Epoxy Composites. In *Journal of Applied Sciences Research* (Vol. 8, Number 7).

Shokri, A., & Sanavi Fard, M. (2022). Corrosion in seawater desalination industry: A critical analysis of impacts and mitigation strategies. *Chemosphere*, *307*, 135640. https://doi.org/10.1016/j.chemosphere.2022.135640

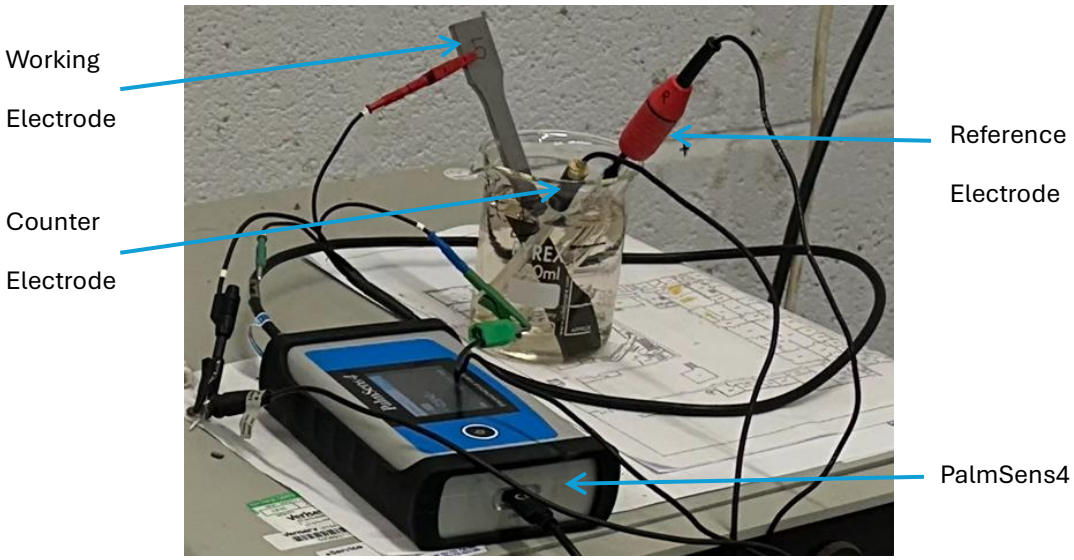
Teymouri, F., Allahkaram, S. R., Shekarchi, M., Azamian, I., & Johari, M. (2021). A comprehensive study on the inhibition behaviour of four carboxylate-based corrosion inhibitors focusing on efficiency drop after the optimum concentration for carbon steel in the simulated concrete pore solution. *Construction and Building Materials*, *296*, 123702. https://doi.org/10.1016/j.conbuildmat.2021.123702

Vallejo Vitaller, A., Angst, U. M., & Elsener, B. (2020). Laboratory tests simulating corrosion in geothermal power plants: influence of service conditions. *Geothermal Energy*, *8*(1), 9. https://doi.org/10.1186/s40517-020-00163-y

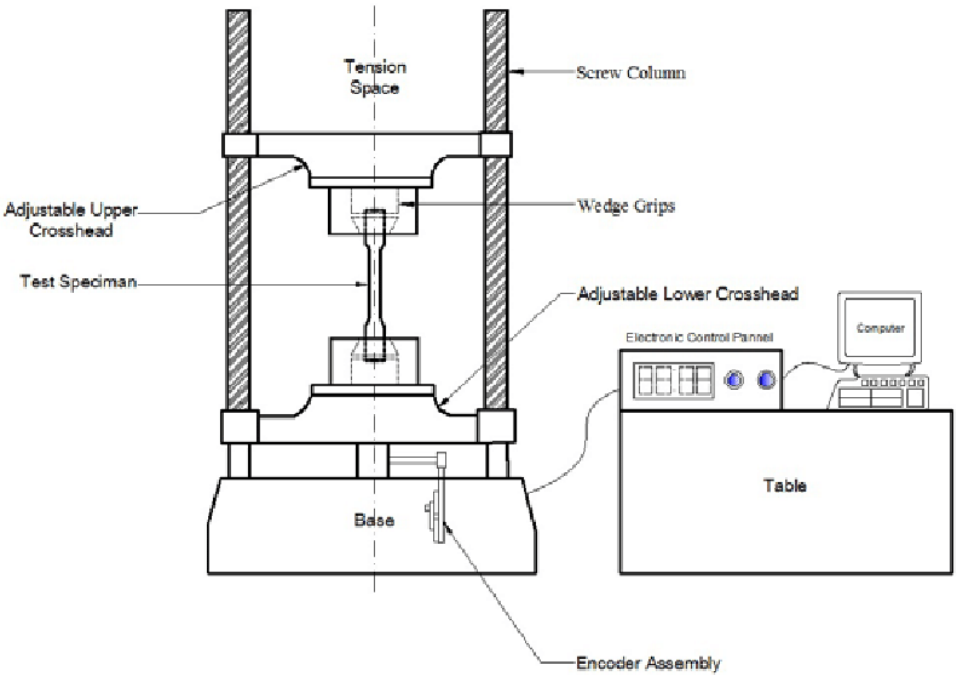
Xu, Y., Zhang, Q., Zhou, Q., Gao, S., Wang, B., Wang, X., & Huang, Y. (2021). Flow accelerated corrosion and erosion–corrosion behavior of marine carbon steel in natural seawater. *Npj Materials Degradation*, *5*(1), 56. https://doi.org/10.1038/s41529-021-00205-1

9. Appendices

Appendix A



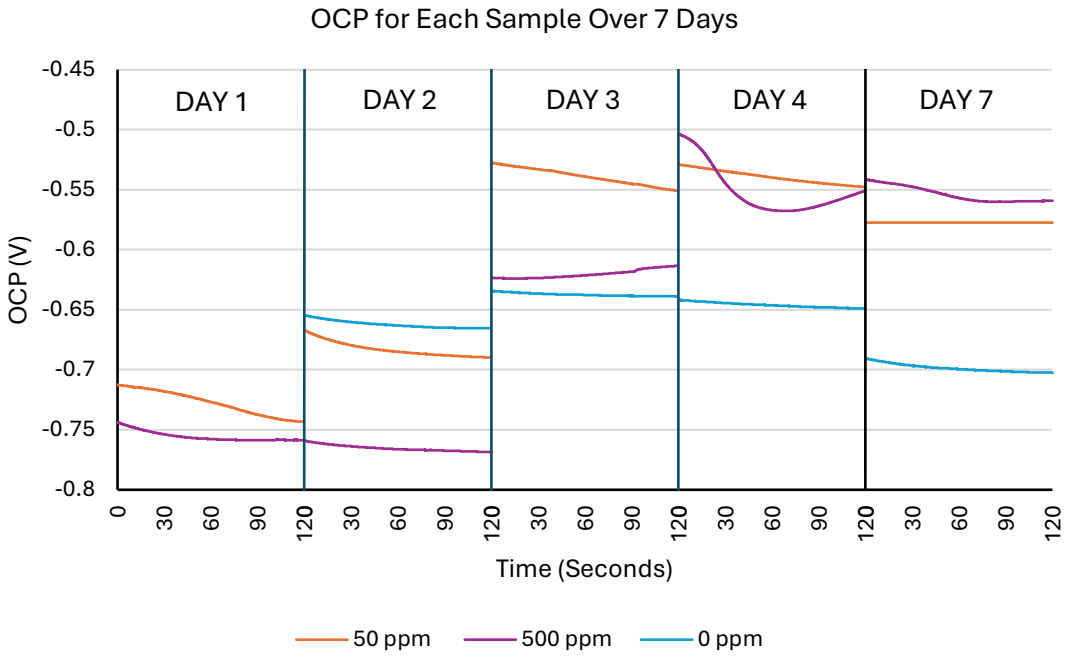
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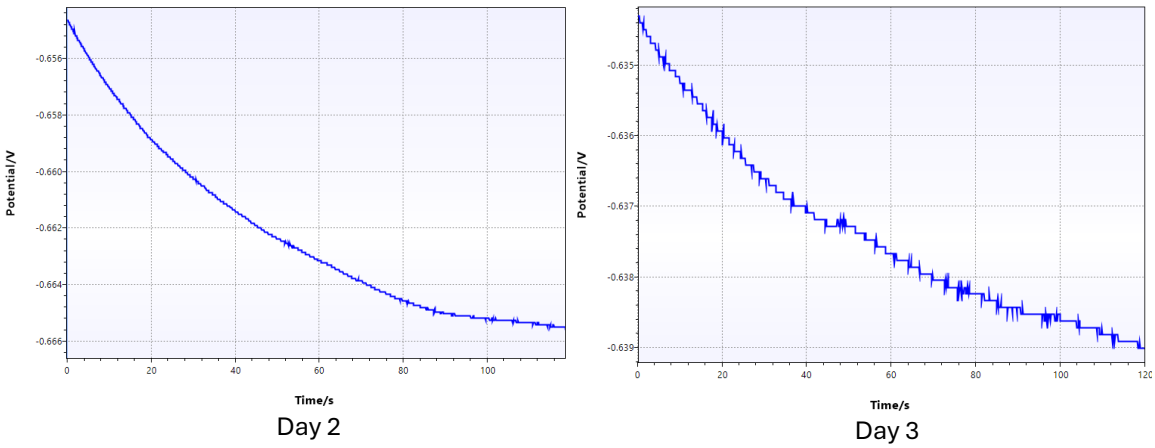
(b)

***Figure A1**: (a) Electrochemical set up, (b) Tensile test set up (Sabarinathan et al., 2012)*

Appendix B



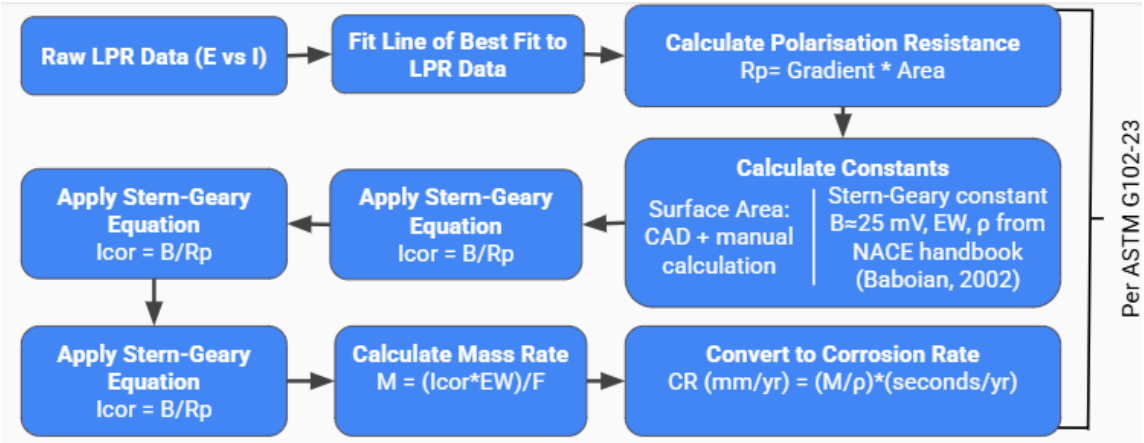
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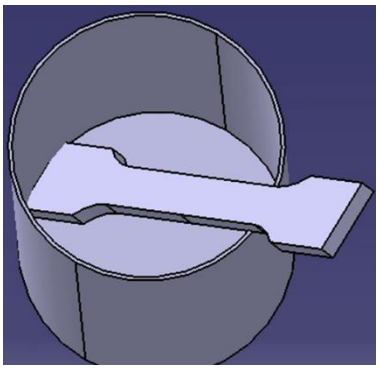
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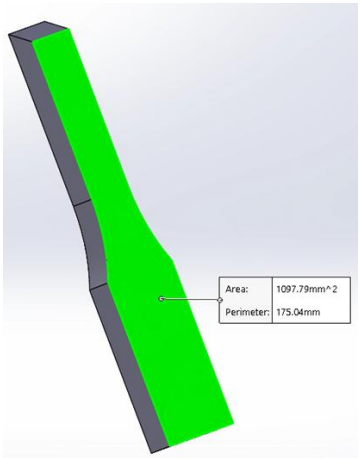
***Figure B1**: OCP for each sample over 7 days, (a) full day 7 exposure, (b) close up of EC noise for 0 ppm on days 2 and 3*

Appendix C



(a)



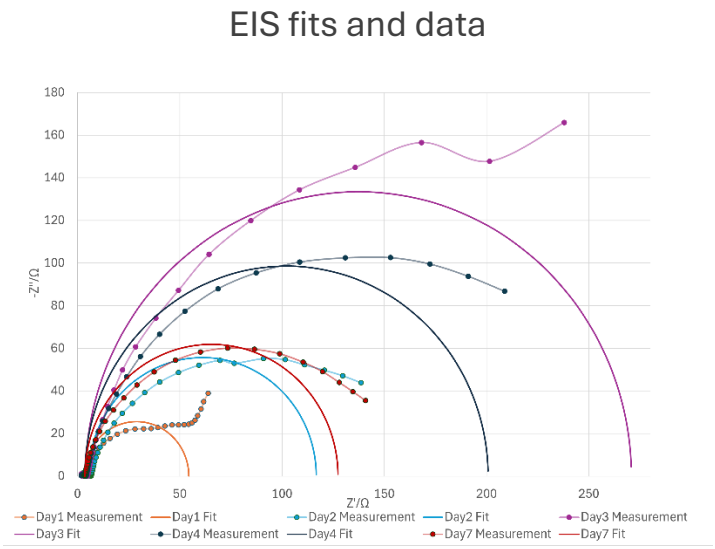


Constrained to rest on beaker realistically, mitigating area overestimation from incorrect mounting angle which would occur without explicitly remodeling the experiment. (specimen cut at liquid level)

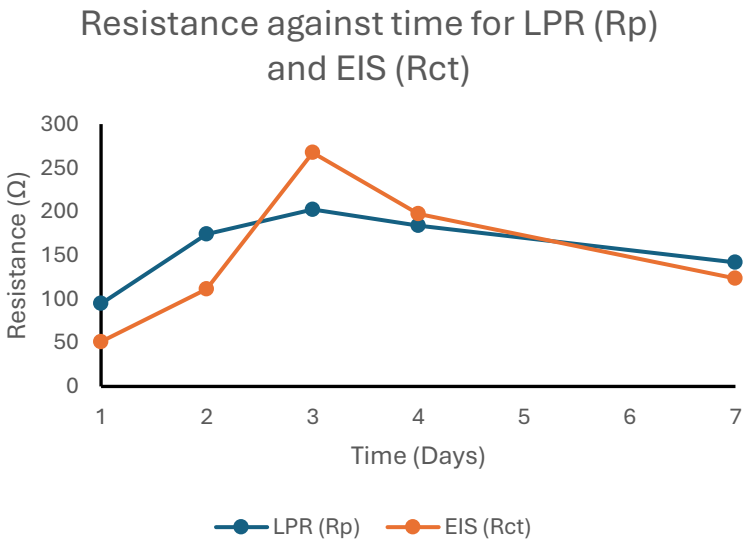
(b)

***Figure C1**: Corrosion rate calculation: (a) General method (G102), (b) Surface area calculation*

Appendix D



(a)

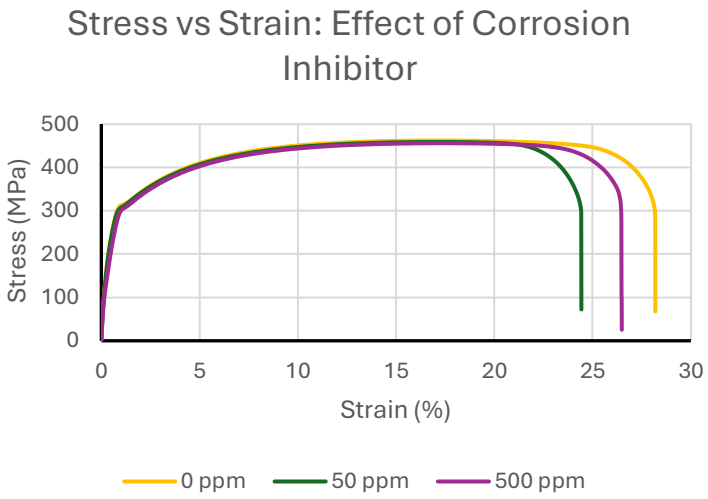


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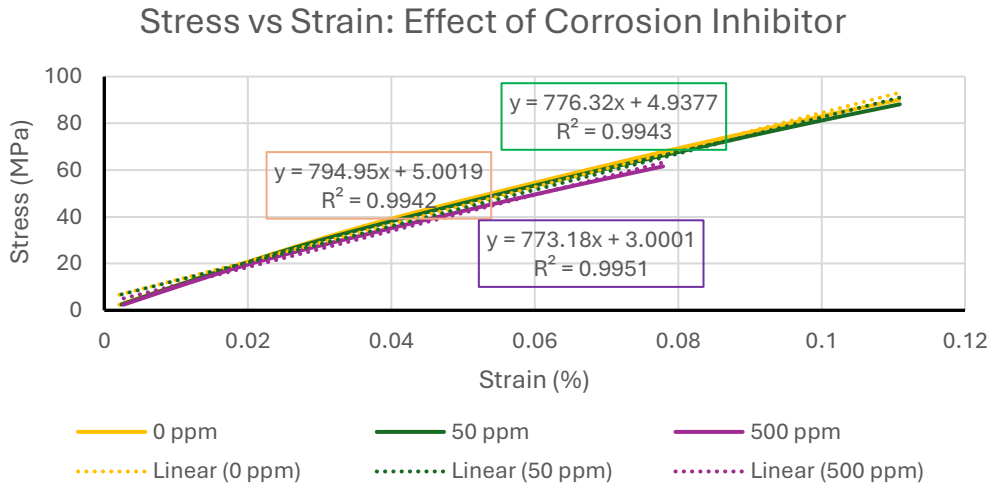
***Figure D1**: EIS Analysis, (a) experimental data and fitted response, (b) EIS and LPR comparison*

The graph shows both EIS and LPR following the same trends both peaking on day 3 and showing degradation afterwards. The observed differences in magnitude is likely due to Rct being obtained from impedance spectroscopy which can include film and diffusion effects (Vallejo Vitaller et al., 2020).

Appendix E



(a)



(b)

***Figure E1**: Tensile calculations, (a) stress strain curves, (b) Close up for young's modulus calculations*

The stress-strain curve was obtained by converting the raw force-displacement data to stress and strain, using the specimen's cross-sectional area for stress and the gauge length for strain. The Young's modulus was calculated through the gradient of the LOBF through the linear elastic region of the stress vs strain graph, whilst ensuring $R^2 > 0.994$.

Sample	UTS (MPa)	Elongation (%)	Young's Modulus (GPa)
0 ppm	462.00	28.19	79.50
50 ppm	458.56	24.43	77.63
500 ppm	455.79	26.48	77.32

Table E1: Tensile test results

Appendix F

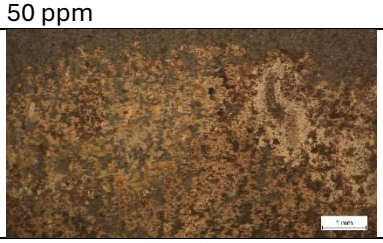
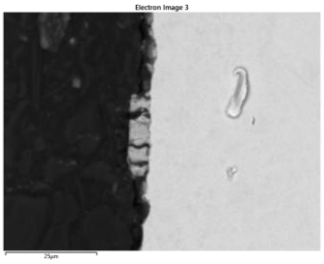
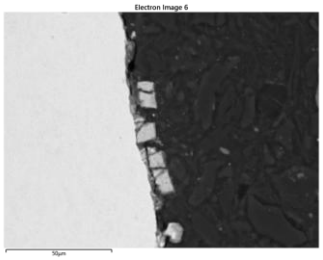
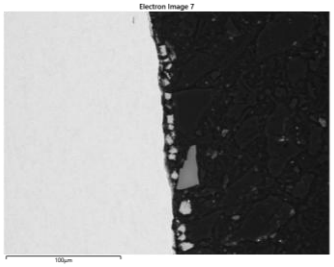
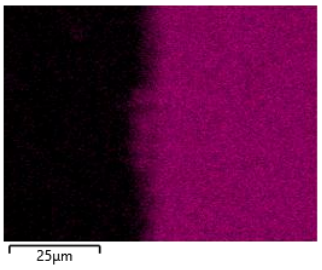
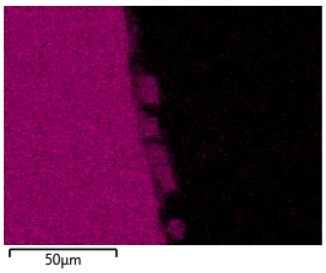
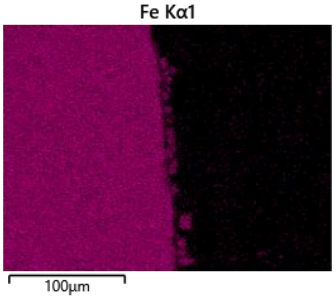
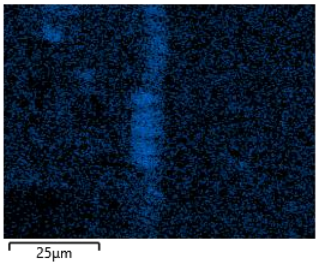
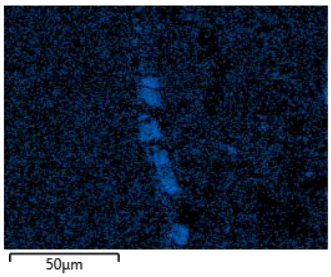
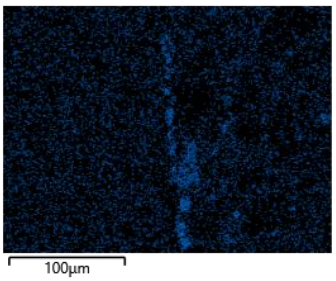
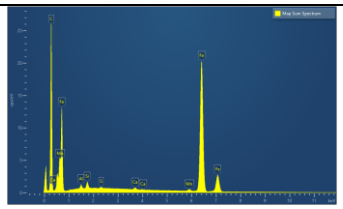
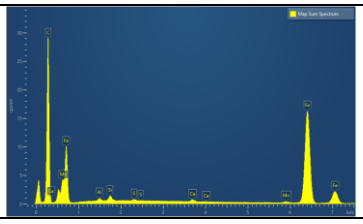
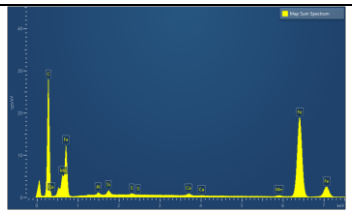
	0 ppm	50 ppm	500 ppm
Macroscopy Images			
BSE Image			
Fe Map			
O Map			
Spectrum			

***Figure F1**: Macroscopy and SEM/EDX images for each sample*

Visual inspection showed the most severe corrosion on the 0 ppm sample with extensive rust formation. A noticeable reduction in visible corrosion was observed at 50 ppm. At 500 ppm corrosion appeared minimal and more uniformly distributed across the surface.

Result Type	Weight %								
Label	C	O	Na	Al	Si	S	Ca	Mn	Fe
Detached material	69.03	21.68		0.50	1.75	0.19	1.17		5.69
Detached material	64.36	18.12	0.22	0.72	1.13	0.21	0.53		14.71
Detached material	41.89	21.37		0.11	0.42	0.14	0.38		35.69
Metal	1.26							0.99	97.75
Epoxy Resin	83.18	14.29	0.14	0.29	0.55		0.71		0.84

Table F1: Point analysis results

Appendix G

1. Hazard Identification			2. Risk Evaluation				3. Risk Control	
No	Hazard	Possible Accident	Existing Risk Control	Severity	Likelihood	Risk Level	Additional Risk Control	Remarks
Lab Experiment Hazards								
1.	NaCl/inhibitor solution handling	Eye/skin irritation	Goggles, gloves, lab coat	2	3	L	Eyewash, spill kit	
2.	Electrochemical testing	Shock/ equipment failure	Training	4	2	M	Dry gloves, PAT test	Highest lab risk
3.	Dog-bone cutting	Laceration	Cut gloves	2	3	M	Sharps bin	Routine prep
Real-World Desalination Pipe Hazards (that cause corrosion)								
1.	Flow velocity	Pipe erosion/ thinning	Material selection	5	5	H	Lab: Flow cell	
2.	Microbially Induced Corrosion/ biofilm	Pitting failure	Biocides	4	4	M	Lab: Inoculated tests	
3.	Galvanic corrosion	Weld failure	Cathodic protection	5	4	H		
4.	Crevice corrosion	Flange leaks	Gasket design	5	4	H		
5.	Chloride driven/general Corrosion	Wall thinning, pitting	Inhibitor dosing, material selection	4	5	H	Regular inhibitor monitoring	

Likelihood / Severity	Rare [1]	Remote [2]	Occasional [3]	Frequent [4]	Almost Certain [5]	Risk Level
Catastrophic [5]	5	10	15	20	25	High (H)
Major [4]	4	8	12	16	20	
Moderate [3]	3	6	9	12	15	Medium (M)
Minor [2]	2	4	6	8	10	
Negligible [1]	1	2	3	4	5	Low (L)

***Figure G1**: Risk assessment*

Hazard	Captured?	Representation	Limitations
Chloride-driven corrosion	Yes	3.5 wt.% NaCl, OCP/LPR/EIS, inhibitor dose response (Appendicies B-D)	Static only; no flow effects
Flow velocity	No	Static beaker	No shear can't assess film erosion
MIC/Biofilm	No	Clean NaCl	No organisms/deposits
Galvanic corrosion	No	Single metal	Isolated electrode
Crevice corrosion	Partially	Gauge geometry	No tight crevices/oxygen gradients

Table G1: Hazard relevance in relation to experiment